

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in chemical/materials process development, DOE-guided optimization, SOP-ready controls, and characterization of polymer and thin-film systems. Experienced translating process variables into material-quality metrics, validation criteria, technical reports, and scale-up specifications for reproducible production transfer.

CORE COMPETENCIES

Chemical Process Development

Polymer/Formulated Materials

DOE/SPC

SOPs & Control Plans

Process Qualification

Root Cause Troubleshooting

Scale-Up Documentation

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, establishing critical process variables, material-quality readouts, reproducible operating windows, and scale-up-ready specifications.
- Convert analytical and thermal-characterization data into SOP-ready controls, technical reports, and troubleshooting logic for process improvement.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed and executed hypothesis-driven experiments across 15+ material variants, diagnosing performance/degradation trends and proposing next-step process or material-design decisions.
- Integrated multi-technique characterization outputs into clear technical conclusions for cross-functional R&D stakeholders.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film coating process development, linking CVD/surface preparation to morphology, interface quality, optical performance, and reproducible process controls.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed and optimized CVD polymer thin-film deposition with DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and coating reproducibility.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, safety, and equipment-tracking systems that improved instrument uptime by 30% and strengthened data quality.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Bio-Based Polymer Process Qualification Current

Defined DOE-guided process windows, analytical readouts, reproducibility criteria, and SOP-ready material-quality specifications for polymer scale-up.
DOE, SPC, TGA/DSC, spectroscopy, process documentation

AFOSR Defense Thin-Film Coatings DoD Funded

Established process-property relationships for CVD/surface-preparation workflows and reproducible high-performance coating conditions.
CVD, QCM-D, surface prep, thin films

Materials Reliability and Failure-Mode Study Chemistry of Materials 2025

Mapped degradation, redox instability, and performance failure modes across functional material variants using orthogonal characterization and stress-test logic.
FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Chemical & Materials Process Development: DOE, process windows, process qualification, validation-style protocols, SOPs, work instructions, technical reports, reproducibility studies

Polymer/Formulated Materials: Bio-based polymers, thin films, coatings, surface functionalization, structure-property-processing relationships, stability-sensitive materials

Characterization & Quality Readouts: FT-IR, UV-Vis-NIR, TGA, DSC, GPC/SEC, LC-MS, GC-MS, NMR, QCM-D, cyclic voltammetry, pH/solution preparation experience

Statistical & Production Controls: SPC, JMP (basic), Python (basic), Excel, control-plan thinking, out-of-trend investigation, root-cause troubleshooting

Cross-Functional Execution: Supplier/process issue communication, lab safety, calibration tracking, equipment coordination, stakeholder reporting

JOURNAL PUBLICATIONS

- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
- Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)